

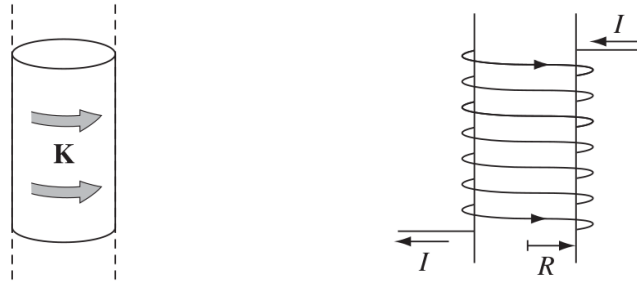
Relativity and Electromagnetism Exercise 3: 2026 Feb 26

Problem 1. From the expression for the magnetic vector potential $\mathbf{A}$,

$$\mathbf{A}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}')}{r} d\tau',$$

obtain the usual Biot-Savart law.

Problem 2. Consider a surface current $\mathbf{K} = K \hat{\phi}$ flowing along the surface of an infinite cylinder of radius R . Calculate the magnetic field for this current configuration.



Problem 3. Assume that the surface current in Problem 1 is produced by a solenoid of finite length L carrying a steady current I and with n closely wound turns per unit length. How is K expressed in terms of the parameters of the solenoid? The solenoid has a finite length while the result in Problem 1 was obtained for an infinite cylinder. Under what conditions, and in which region, can the result of Problem 1 be applied to a finite solenoid?

Problem 4. Inside of a *perfect conductor* the electric field $\mathbf{E} = 0$, and any net charge resides only on the surface.

- a. Show that the magnetic field is constant, $\partial \mathbf{B} / \partial t = 0$, inside a perfect conductor.
- b. Show that the magnetic flux through a perfectly conducting loop is constant.

Problem 5. A *superconductor* is a perfect conductor with the additional property that the magnetic field $\mathbf{B}$ inside it is not just constant, but *zero*. This property is known as *flux exclusion*.

- a. Show that the current in a superconductor is confined to the surface.
- b. Most materials are superconducting only *below* a certain critical temperature T_c . Consider a superconducting material above its critical temperature in a uniform magnetic field $\mathbf{B} = B \hat{z}$. For simplicity, assume that this is the magnetic field both outside and inside. What happens as the temperature is lowered to below the critical temperature?
- c. Calculate the induced surface current distribution on the superconductor assuming that it is a cylinder placed along the z direction. You can use your answer from Problem 1.

The physics described here is known as *flux expulsion* or the *Meissner effect*.